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Information Presentation Response

After learning about Weapons of Math Destruction, my perspective on data-driven decision-making has changed, particularly regarding how algorithms reinforce societal inequalities. Previously, I believed that data-driven systems were inherently objective, but the book highlights how algorithms embed biases, such as in predictive policing and hiring discrimination (O’Neil, 2016). Additionally, ProPublica’s research shows that risk assessment algorithms in the criminal justice system disproportionately classify Black defendants as “high risk” (Angwin et al., 2016). If future developments ensure greater transparency in algorithm design and stronger regulatory oversight, my views on algorithmic fairness may evolve.

Algorithmic University Admissions

Introduction

Developing the Program to Optimize Student Tuition (POST) algorithm requires careful consideration. Including biased or problematic data can lead to discrimination. While machine learning can help identify candidates, it is crucial to exclude data that may introduce or amplify bias.

Excluding Race and Gender

One important step is to exclude race and gender from the POST algorithm. Including these factors can result in discrimination. For example, Amazon’s hiring algorithm

discriminated against women because of biases in the training data (Dastin, 2018). Machine learning models trained on biased data tend to repeat existing inequalities (O'Neil, 2016). To ensure fairness, race and gender should not be used in admissions.

Limiting Socioeconomic Indicators

Socioeconomic indicators, like family income and parents' education, should be minimized. These factors can unfairly benefit students from wealthier backgrounds. Research shows that using socioeconomic data can actually increase disparities (Clark, 2023). In criminal justice, algorithms have labeled low-income individuals as high risk, reflecting bias (Angwin et al., 2016). Applying similar methods in university admissions would disadvantage lower-income students.

Excluding Criminal History

The model should also exclude criminal history. Including this data can disproportionately affect certain racial and socioeconomic groups. Studies have shown that risk assessment tools often classify Black defendants as high risk, while underestimating white defendants (Angwin et al., 2016). Using criminal history in admissions would unfairly penalize some applicants based on systemic biases.

Avoiding Geographic and Taxation Data

Address and taxation history should also be omitted. These factors can serve as indirect indicators of race and income. Mortgage lending algorithms have discriminated against minority neighborhoods for similar reasons (Clark, 2023). Including address data could unfairly favor applicants from wealthier areas.

Focusing on Academic and Extracurricular Achievements

A fair admissions system should focus on academic achievements and extracurricular activities. These metrics reflect a student's potential more accurately. They do not introduce bias related to demographics. Without proper safeguards, the POST algorithm risks becoming a tool of unfairness rather than equity (O'Neil, 2016).

Conclusion

To develop a fair POST algorithm, it is important to exclude biased data. Race, gender, socioeconomic status, criminal history, and geographic data should not be used as predictors. Instead, admissions should focus on merit-based criteria. This will help ensure that the process is fair and equitable.

Additional Perspectives and Future Considerations

It is important to consider diverse perspectives. Some researchers argue that using socioeconomic factors can help address educational inequalities (Espinoza et al., 2019). Including these viewpoints will help build a more balanced argument.

Common Crawl

Introduction

Common Crawl's large-scale web scraping offers diverse training data for AI models. However, its data selection and filtering can introduce biases. These biases can shape AI models in harmful ways. They may reinforce dominant narratives, ignore underrepresented voices, or spread misinformation.

Bias in Data Selection and Filtering

One major issue is bias in data selection. Common Crawl prioritizes high-traffic, frequently updated websites. This results in an overrepresentation of mainstream sources.

Niche or minority perspectives often get overlooked (Dodge et al., 2021). Popular news outlets, Wikipedia, and corporate sites dominate the data. This can reinforce Western-centric narratives (Bender et al., 2021).

Additionally, Common Crawl does not check for reliability. As a result, misinformation can persist. This can lower the accuracy of AI models (Brown et al., 2020).

Bias in Language and Culture

Another problem is language bias. Common Crawl contains mostly English-language content. It has few non-English sources (Brown et al., 2020). AI models trained on such data become fluent in English. But they may struggle with less common languages. This leads to poor-quality responses for users from non-English regions. They may encounter grammatical errors or culturally insensitive outputs.

Additionally, the heavy focus on Western sources shapes AI content toward Eurocentric viewpoints. This limits cross-cultural representation (Bender et al., 2021).

Harmful Stereotypes and Discrimination

Common Crawl also includes unfiltered social media posts and forum discussions. These often contain biased language related to gender, race, and socioeconomic status. AI models trained on such data may repeat or even amplify these biases (Dodge et al., 2021). For example, biased data led to hiring algorithms that unfairly disadvantaged women (Dastin, 2018).

Similarly, biased law enforcement data has led to racial profiling in predictive policing. This has resulted in unequal treatment of minority groups (Angwin et al., 2016). Without filtering and bias mitigation, AI risks reinforcing harmful stereotypes.

Issues with Content Verification

Common Crawl does not verify the sources it scrapes. As a result, it may include unreliable blogs, misleading opinions, or conspiracy theories (Brown et al., 2020). This makes AI models prone to hallucination—generating inaccurate or misleading information. If AI cannot tell fact from opinion, it may spread misinformation. This is risky in areas like healthcare, law, or politics.

Mitigating Bias in AI Models

To reduce bias, developers should improve data filtering. They need to prioritize reliable and diverse sources. Including more non-English content would improve global representation. Applying bias detection algorithms can help find harmful patterns (Dodge et al., 2021).

Additionally, making data usage more transparent would promote fairness. This includes documenting data sources and selection methods (Bender et al., 2021).

Conclusion

Common Crawl is useful for AI training, but it has biases. Overrepresentation of English and mainstream media can skew AI content. Unverified data can spread misinformation. To build ethical AI systems, developers must address these issues. Improving data diversity and reliability is essential.

Reference

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